

ROHAN PODURI

Curriculum Vitae

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EDUCATION

University of Wisconsin–Madison

Bachelor of Science in **Economics**; Minors: **Data Science, Development Economics**

Expected Graduation: May 2026; **GPA:** 3.73 / 4.0 (*Dean's List*)

Relevant Coursework: Econometric Modeling, Economic Forecasting, Data Science Modeling and Programming, Economic Data Analysis, Data Visualization, Machine Learning, Industrial Organization, Consumer Finance, Behavioral Economics, Development Economics, Growth Economics

TECHNICAL SKILLS

Econometric Methods: Regression analysis, panel data methods (fixed effects, random effects), regression discontinuity design, instrumental variables, difference-in-differences, time-series models (ARIMA, VAR), machine learning (ensemble methods, regularization, cross-validation)

Programming Languages and Tools: Python (pandas, NumPy, matplotlib, scikit-learn, statsmodels), R (tidyverse, ggplot2, caret, forecast), Stata (data cleaning, regression, panel methods), SQL, Excel, Tableau, PowerPoint, GitHub (Version Control)

Research and Communication: Literature synthesis, empirical research design, statistical reporting, technical writing, exhibit preparation, interactive dashboards (Tableau), geographic visualization with interactive filters, income quartile groupings, visual communication, panel data construction, crosswalk creation, data merging, quality assurance, reproducible workflows

RESEARCH EXPERIENCE

Research Assistant

Department of Applied Economics, University of Wisconsin–Madison

Principal Investigators: Dr. Priya Mukherjee, Dr. Nishith Prakash

February 2025 – Present

- Supporting empirical research in **political economy, development economics, and causal inference** through data engineering, statistical analysis, and methodological foci.
- Implemented **vision-based machine learning models (VLM)** on large-scale administrative datasets to automate data extraction and validation, achieving **near-zero classification accuracy**, substantially reducing manual coding time and improving data consistency across multiple research projects.
- Constructed the **first-ever longitudinal state-district crosswalk spanning 70+ years** of Indian administrative data. Resolved **boundary changes, district renamings, splits, and missing records** to enable consistent **panel analysis** for political economy research.
- Cleaned, merged, and validated multi-year administrative datasets using **Python, R, Stata, and Excel**. Implemented quality checks, created documentation, and ensured **reproducibility of data workflows**.
- Supported **causal identification strategies** including **regression discontinuity designs, fixed effects estimation, and instrumental variables approaches**. Produced robustness checks, and diagnostic plots for faculty review.

PROFESSIONAL EXPERIENCE

Economic Event Analyst

SwingPhi (Fintech Startup)

July 2025 – November 2025

- Conducted **event-driven economic analysis** around **earnings releases and macro announcements**, linking market responses to economic fundamentals.
- Built **reproducible analytical workflows** in **Python** (pandas, NumPy) and **Excel** for peer comparisons, trend decomposition, and benchmarking.
- Synthesized quantitative findings into research briefs for **internal and public-facing audiences**, producing data driven analyses that averaged **250+** external viewers, supporting informed decision-making and client engagement.

Market Research Intern

Podigit Pvt. Ltd. (E-commerce Consulting)

December 2024 – January 2025

- Developed **cost and revenue optimization models in R and Excel** to evaluate operational strategies for e-commerce fulfillment. Identified adjustments that **reduced projected costs by 22% and increased margins by 15%**.
- Evaluated pricing and fulfillment strategies (**FBA vs. FBM**) across multiple scenarios. Demonstrated **profitability differences up to 60%** depending on product characteristics, supporting a **projected 25% profit increase**.
- Conducted **competitor benchmarking and market analysis**, translating quantitative findings into **client-ready recommendations and visual dashboards**.

RESEARCH PROJECTS

1. Data Visualization: College Mobility, Innovation, and Economic Outcomes (Tableau)

Research Question: How does patent production vary by **institutional selectivity, type, and parental income**, and where are high-patent institutions geographically concentrated?

Data: College-level data including patent output, selectivity tiers, institutional types, median parental income, adult earnings, and geographic coordinates.

Methods: Built interactive **Tableau dashboards with boxplots, bar charts, scatterplots, histograms, and county-level polygon maps** with interactive region filters. Created income quartile groupings to analyze socioeconomic patterns.

Findings: Patent production is **highly concentrated in Ivy Plus** and **highly selective** institutions. Private nonprofits outperform public colleges. **Higher parental income correlates with greater patent output**. Geographic clustering is evident in the Northeast and West. Adult earnings vary sharply by selectivity tier.

Contribution: Demonstrated proficiency in **Tableau dashboard design, geographic mapping with interactive filters, and data-driven storytelling** for policy-relevant research on higher education innovation and economic mobility.

2. Forecasting U.S. Consumer Credit: A Trend-Seasonal AR(1) Model

Research Question: Can a parsimonious time-series model accurately forecast total consumer credit outstanding **12 months ahead**, and how well calibrated are its forecast intervals?

Data: Federal Reserve G.19 Consumer Credit release (**TOTALSL series**), monthly data through October 2025.

Methods: Estimated **AR(1) model in logarithms** with **deterministic trend and monthly seasonal indicators**. Constructed **95% forecast intervals**.

Findings: Point forecast: **\$5.15 trillion**. Forecast interval: **[\$5.00T, \$5.50T]**. Realized: **\$5.08 trillion**. Forecast error: **-\$70 billion (1.4% deviation)**.

Contribution: Demonstrated that **simple trend-seasonal models produce well-calibrated forecasts** for persistent macroeconomic series.

3. Suicide Trends in Los Angeles County (Undergraduate Thesis)

Advisor: Dr. Priya Mukherjee | *December 2025 – Present*

Research Question: Did stricter state-level COVID-19 policy restrictions causally affect suicide mortality rates in the United States?

Data: **CDC WONDER** state-level monthly mortality data (ICD-10 suicide codes, 2014-2023), **Oxford COVID-19 Government Response Tracker** (policy stringency index), **BLS** unemployment and income data, Census population estimates. Preliminary descriptive work used LA County data.

Methods: Implementing **staggered difference-in-differences (DiD)** to exploit variation in timing and intensity of state COVID-19 policy adoption. Deploying **triple-differences (DDD)** to test differential effects on young adults (18-25) vs. older adults (45-65). Testing unemployment mediation to identify economic channels.

Status: **In progress**. Constructing state-level panel dataset integrating **mortality, policy stringency, economic indicators, and demographic controls**. Moving from descriptive LA County analysis to **quasi-experimental causal identification** using recent advances in DiD methodology.